

Response to Reviewers

Paper 1217, Depth, Breadth, and Bias: Structural Diffusion of Political Content on Divergent Platforms

We thank the reviewers and the SPC for their careful and constructive comments. We found the feedback very helpful in identifying several places where the original manuscript needed clearer definitions, additional validation, and more cautious interpretation. We have revised the manuscript substantially in response. All additions and major revisions are highlighted in blue in the revised manuscript.

The main changes include additional robustness checks for repost-cascade reconstruction, expanded validation of the ideology labels, clarification of the post-level unit of analysis, sensitivity analyses using alternative cascade and ideology specifications, and a revised regression specification after identifying a problem with the original scale estimator. We also expanded the descriptive reporting, motif analysis, and discussion of the scope and limitations of the findings.

Across these sensitivity analyses, estimates sometimes change substantially in magnitude, but none overturns the central conclusions. The depth difference persists across alternative samples and cascade representations, while the user-collapsed analysis shows that part of the post-level breadth difference reflects repeat participation. The association with ideological composition and interaction alignment remains substantial across alternative ideology thresholds, label-noise simulations, and composition specifications. Some checks also narrow the interpretation of particular findings. For example, collapsing repeated posts by the same user reduces the breadth difference, indicating that repeat participation contributes to the post-level breadth contrast, while the partisan-only analysis shows that the residual center category contributes to part of the statistical attenuation. We have revised the manuscript to make these qualifications explicit.

The first seven items are the seven points raised in the SPC decision, each answered once. Where a reviewer raised the same issue it is listed in the Raised by column and answered there rather than repeated. Items C8 onward are the remaining reviewer-specific comments.

I worry that the rest of the paragraph is the wrong level of detail for this stage. Coming back to this after many months (which is the case for the reviewers), it is hard to know exactly what you are saying here. I would suggest that you cut the parts after "but none overturns the central conclusions" and say you added qualifications to the text when necessary.

#	Raised by	Reviewer point	Response	Where
C1	SPC-1, R1-R3	Repost cascades are reconstructed rather than directly observed, so the	We addressed this in two ways. First, we rebuilt all 244,129 repost cascades under alternative linking rules, together with a 40-draw random-linking specification. Across these, the platform-by-size interaction stays within $[-0.012, +0.031]$ for breadth, $[-0.028, +0.028]$ for depth, against $+0.170$ and -0.182 for the reply cascades. Second, because follower edges are in a post-collection snapshot without edge creation, we simulated timing error by randomly removing 5% of follower edges and reconstructing the cascades; the platform interactions remain much smaller than the timing values. Both follower networks were collected during the same period, although we do not assume the timing error is the same across platforms. We also now refer to these cascades as reconstructed throughout.	App. D, Fig. 9
C2	SPC-2, R2, R3	It was unclear whether cascade nodes represent users or posts, are	We have clarified that reply-cascade nodes are posts, not unique users. A user who replies multiple times therefore contributes multiple nodes carrying the same user-level ideology label. We also reconstructed the reply cascades after collapsing repeated appearances by the same user. The depth difference remains substantially smaller. We now state explicitly that part of the post-level breadth contrast reflects repeat participation within threads.	Methods; App. E
C3	SPC-3, R1, R2	The ideology validation was limited, the Center category may combine moderate with uncertain classifications, and the manuscript should show robustness to plausible label error.	We addressed these concerns together because they all bear on the measurement of ideology. We now report class-specific and platform-specific precision and recall with bootstrap intervals on the 171 cases where both annotators agreed. Accuracy against consensus is 0.86, inter-annotator kappa is 0.77, and model kappa is 0.61 and 0.74. We also define the Center category explicitly as a residual and recompute composition and alignment using partisan users only, where the association remains substantial although smaller. Finally, we propagate label error through Model 3c under four error processes, with 100 draws per scenario. Median attenuation ranges from 70 to 86 percent, against 84 and 86 percent with unperturbed labels, and even the least favorable draw retains 66 percent. The pooled-error scenario produces the largest reduction; the platform-specific and bootstrap scenarios, closest to the error we measured, stay nearest the unperturbed estimates.	App. B, Table 3

I find it hard to conclude what your analysis for this first point explains. does this mean it is robust? doesnt sound like it. also 5-30% deletion is really a lot. If we think timing is the problem, given overall stability of networks, i would be surprised you would be 30% off. what am I getting wrong here?

what do you mean by platform-by-size? it seems these estimates are crossing the zero line so what does that mean? why do you have a range for the first two but not the last two? it sounds like you are comparing depth measures to "correspondin reply cascade", not sure what that means

how do you collapse the repeated appearance by the same user? what if a responds to be who responds to c who responds to a? is there a back edge from a to c?

why two numbers for human to model kappa? for the two raters?

association with what remains smaller?

ranges of what?

again, i am getting lost here. reduction in what? how much alignment explains?

#	Raised by	Reviewer point	Response	Where
C4	SPC-4, R2, R3	Alignment is calculated from the same reply edges that determine cascade structure, making causal language inappropriate.	We agree and have revised the manuscript throughout to avoid a causal interpretation of the alignment results. Alignment and cascade geometry are jointly realized features of the same conversation, so the observed association does not establish a direction of influence. The Results now state this limitation directly, and the Discussion consistently describes composition and alignment as statistically associated with the platform difference rather than as causal mechanisms.	Results; Limitations
C5	SPC-5, R1	Descriptive reporting is thin: cascade counts by size, raw motif counts, whether the null model handles motif overlap, and topic-model detail.	We expanded the reporting throughout. Cascade counts and size distributions are now given by platform and cascade type; this also made clear that cascades are common and far more frequent on Bluesky than on Twitter. The null model carry no variation in breadth or depth we excluded them, where the baseline platform model does but present and the combined ideology specification still accounts for most of it. We added observed counts for all 54 ideology-labeled three-node motifs on both platforms. We describe the randomization and counting procedure explicitly: the same overlapping-instance enumeration is applied to observed and randomized graphs alike, so overlap is treated consistently, and because motifs share substructure we interpret the results as patterns across motif families rather than as independent evidence. We also report the topic-model corpus size (125,623 root posts), the outlier reduction procedure and settings, the remaining outlier share, and the eleven topic categories.	it is odd to me that you excluded something that is same across platforms and somehow the differences became smaller. again, i might be misunderstanding what you are saying here.
C6	SPC-6, R2	The main text describes robust regression with Huber loss, whereas the appendix referred to OLS with HC3 standard errors.	We corrected this inconsistency. The main reply-cascade models use the Huber loss. Reviewing the procedure, we found the default median-absolute-deviation scale degenerates because of the large mass of root-only cascades, so the main models are refit with Huber’s Proposal 2 scale. Table 1 now reports the corresponding estimates and standard errors, and Appendix I reports sensitivity results across samples and scale estimators. OLS with HC3 errors is used only for the repost-reconstruction robustness analysis, where the Huber scale estimate also degenerates.	Methods; App. I, Table 6

#	Raised by	Reviewer point	Response	Where
C7	SPC-7, R1-R3	The paper examines one month of Biden- and Trump-related discussion during an unusual election period, in a single national context. It should clarify what generalizes and what the two cascade shapes imply.	We revised the Discussion to state both the substantive implications and the limits of the comparison, and gathered the limitations into a single section. We distinguish the structural interpretation of broad and shallow from narrow and deep reply cascades from any normative evaluation of those forms, and make clear that the observed ideological composition and absolute cascade statistics are specific to the candidate-centered sampling frame and period studied here. We also state explicitly that both platforms are U.S.-centered and the sampled conversation is American, so the patterns describe a single national context, and that comparable analyses elsewhere are needed to establish how broadly they generalize.	Discussion; Limitations
C8	R1	The estimated left share on Truth Social and right share on Bluesky appear high relative to previous estimates.	These estimates are not directly comparable to prior platform-wide prevalence estimates because the studies differ in both sampling frame and how ideology is measured. Our labels summarize expressed political position within the sampled Biden and Trump discourse rather than a context-independent ideology, and a position that appears left-leaning in a Trump-centered conversation need not imply a consistently left-wing position across candidates or issues. We therefore do not treat exact agreement with a particular platform-wide percentage as the validation criterion. Instead, we evaluate whether the labels are interpretable and sufficiently reliable for the comparative analysis in which they are used. We report human validation by class and platform, quantify the direction and uncertainty of classification error, apply a prevalence correction, and propagate measured label error through the downstream models. The correction reduces the estimated Bluesky right share from 24.2% to 17.9% and the Truth Social left share from 11.5% to 1.9%.	App. B; Limitations
C9	R1	The rationale for examining highly followed users only on Truth Social was not sufficiently developed, and a symmetric top-n rule might be preferable.	We expanded the justification for treating the unusually high-follower Truth Social accounts separately. The upper tail is far more concentrated on Truth Social: the largest account has 56 times the follower count of that platform’s 99.9th percentile account, against a ratio of 5 on Bluesky. We therefore treat this as a platform-specific concentration rather than imposing the same numerical cutoff on Bluesky, where the upper tail shows no comparable separation.	App. A

#	Raised by	Reviewer point	Response	Where
C10	R1	How much of the observed reply depth consists of back-and-forth interaction between the same two users?	We added a direct analysis of A -> B -> A exchanges. Among cascades with depth of at least two, 68.8% on Bluesky and 70.0% on Truth Social contain at least one such exchange. These exchanges account for 20.0% and 15.6% of reply edges, respectively. The similar prevalence across the two platforms suggests that the observed depth difference is not primarily due to one platform containing more dyadic back-and-forth conversations.	App. F
C11	R1	The alignment results were difficult to follow.	We revised the alignment subsection to provide a clearer definition, define the notation at first use, and include a worked example showing how the measure is calculated within a reply tree.	Methods
C12	R1	Several figures were difficult to read because of overlapping labels and unclear plotted quantities.	We revised the figures to improve readability, including moving motif-family labels outside the plotting area and correcting labels and annotations that extended beyond the axes. The revised figures are rendered at 300 dpi and no longer contain overlapping elements.	Figs. 1, 4, 5, 7, 11
C13	R1	The literature review would benefit from more recent information-spreading work.	We updated the Related Work section with more recent work on cascade structure, information diffusion, and newer social media platforms, including recent work on Bluesky.	Related Work
C14	R1	Some notation was unexplained and formatting was inconsistent.	We reviewed the manuscript for undefined notation, inconsistent terminology, and formatting issues and corrected these throughout.	Throughout

We thank the reviewers and the SPC again for their careful and constructive feedback. **Regardless of the final decision,** we feel that this revision has made the project substantially stronger and more robust. The comments pushed us to clarify several parts of the analysis, add sensitivity checks that we had not originally included, and narrow some of the interpretations where appropriate. We very much appreciate the time the reviewers spent on the paper and the opportunity to revise it.